Artificial Intelligence Across the Brain Tumour Care Pathway: A Narrative Review of Imaging, Computational Pathology, Data-Efficient and Federated Learning, and Clinical Translation

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Abstract

Artificial intelligence (AI) for brain tumours has moved from single-task proof-of-concept models to large benchmarks, foundation models, multi-institutional federations and the first prospective reader studies. This narrative review synthesises that evidence along the clinical pathway, from pre-operative magnetic resonance imaging (MRI) through tissue diagnosis to surgery and follow-up. It draws on a structured evidence base of open-access studies published mainly between 2019 and 2026, each read in full and recorded with its design, data, method, results and limitations. Five findings stand out. First, automated tumour segmentation on curated multi-sequence MRI is technically mature, and well-configured convolutional baselines remain hard to beat; the open problems are incomplete protocols, under-represented populations and post-operative scans. Second, imaging-based prediction of molecular markers is promising for IDH status but inconsistent for other markers, and simple clinical models are sometimes as good. Third, computational pathology has shifted to slide-level weak supervision and foundation encoders, with neuropathologist-level performance for adult diffuse gliomas and fast intra-operative classification, although site-specific signatures persist. Fourth, label-efficient and federated learning make multi-institutional models possible without moving patient data, but heterogeneity between small sites remains a central difficulty. Fifth, the benefit of AI to clinicians is uneven: assistance improves average reading time and accuracy, yet incorrect suggestions mislead readers and common explanation methods rarely meet clinical needs. We conclude that progress now depends less on new architectures than on representative data, honest external validation, robustness to real-world incompleteness and evaluation of clinicians and AI working together.

Keywords: brain tumour; glioma; deep learning; MRI segmentation; computational pathology; federated learning; human-AI collaboration; neuro-oncology

1. Introduction

Brain and central nervous system tumours are uncommon but carry high mortality and morbidity, and their care depends on a chain of specialist judgements: detecting and outlining the lesion on MRI, establishing an integrated diagnosis that combines tissue appearance with molecular markers, planning a resection that preserves function, and monitoring the patient over time (Khalighi et al., 2024; Baker et al., 2024). Each step draws on scarce expertise in neuroradiology, neuropathology and neurosurgery, and each has attracted a growing body of AI research.

The field has changed quickly. Public challenges such as the Brain Tumor Segmentation (BraTS) benchmarks created shared datasets and metrics (Baid et al., 2021); self-configuring pipelines made strong segmentation baselines widely available (Isensee et al., 2021); computational pathology moved from patch-level to slide-level learning and then to large foundation models (Lu et al., 2021; Chen et al., 2024c); and federated learning showed that models can be trained across dozens of institutions without centralising patient data (Pati et al., 2022). At the same time, reviews of AI in neurosurgery and neuro-oncology consistently report the same barriers: limited and unrepresentative data, weak external validation, poor interpretability and uncertain clinical integration (Awuah et al., 2024; Tangsrivimol et al., 2023; Iqbal et al., 2022; Agadi et al., 2023).

Existing reviews tend to cover one modality or one technique. This review instead follows the patient pathway and connects four strands that are usually discussed separately: image analysis, tissue analysis, learning under data scarcity and privacy constraints, and the evidence on how clinicians actually use AI. Our aim is to describe what is established, what remains open, and which lessons recur across strands, with particular attention to settings where data are scarce and heterogeneous.

The question matters most where specialists are few. Neurosurgery depends on technical precision, and about a quarter of its adverse events have been attributed to preventable technical error (Sugiyama et al., 2025). Neurosurgeons worldwide report strong interest in AI but limited formal knowledge of it (Boaro et al., 2025), and the public datasets on which most models are built leave many populations and subspecialties unrepresented (Chan et al., 2025). Understanding which results transfer to routine care in other settings, and which do not, is therefore as important as tracking the newest methods.

The review is organised as follows. Section 2 describes how the evidence was assembled. Section 3 summarises the public data resources that shape the field. Sections 4 to 9 follow the pathway: imaging, computational pathology, integration of evidence, learning with limited and shifting data, federated learning and the reliability of model outputs. Section 10 turns to clinical translation, including surgery, clinician-AI interaction and language models. Section 11 discusses how such systems should be evaluated, and Section 12 draws the strands together.

2. Approach of this review

This is a narrative review. It is based on a structured evidence base of open-access studies assembled by the author between October and November 2026 from arXiv, PubMed Central and publisher open-access collections, using search terms covering brain tumour imaging, glioma, computational pathology, data-efficient learning, federated learning, uncertainty and human-AI interaction. Studies from 2021 onwards were prioritised, with earlier landmark papers included where they define a method or benchmark in current use. Papers without an accessible full text and publications from predatory venues were excluded.

Every included paper was read in full and recorded in a common template: aim, study design, datasets and sample size, preprocessing and data split, model details, training settings, evaluation protocol and metrics, baselines, key quantitative results, statistical analysis, code and data availability, and stated limitations. The studies cited here were selected from that evidence base for their relevance to the brain tumour pathway or, for human-AI interaction, for the strength of their design. Because the review is narrative, no pooled effect sizes are computed; quantitative results are reported as published, and the main limitation of each line of evidence is noted alongside it. Table 1 summarises the stages of the pathway covered and the state of the evidence for each.

Table 1. Stages of the brain tumour pathway covered by this review, and the state of the AI evidence for each.

Pathway stage	Main AI tasks	State of the evidence	Main open problem
Pre-operative MRI	Tumour segmentation, metastasis detection, molecular marker prediction	Mature on curated data; molecular prediction mixed	Missing sequences, under-represented populations, external validation
Tissue diagnosis	Slide-level classification and grading, mutation prediction, intra-operative diagnosis	Specialist-level for major glioma types	Site-specific signatures, paediatric and rare tumours
Integrated diagnosis	Fusion of radiology, pathology and molecular data	Promising on complete cohorts	Robustness to missing or poor-quality sources

Pathway stage	Main AI tasks	State of the evidence	Main open problem
Learning across hospitals	Label-efficient, federated and privacy-preserving training	Feasible at large scale	Small heterogeneous sites, privacy leakage
Surgery and follow-up	Extent of resection, tractography, planning, intra-operative imaging	Early; mostly geometric accuracy	Surgeon acceptability and patient outcomes
Clinician use	Reader assistance, language-model decision aids	Average benefit in reader studies	Automation bias, variability between clinicians, explanations

3. Data resources and benchmarks

Shared datasets have shaped what brain tumour AI can do and how it is judged. Table 2 lists the resources cited most often in this review. The BraTS adult glioma benchmark is the reference point for segmentation, but it is curated and pre-processed with a standard pipeline, which routine clinical scans are not (Baid et al., 2021). Newer challenges have widened the scope to paediatric tumours (Kazerooni et al., 2024), meningioma radiotherapy planning (LaBella et al., 2026) and patients from Sub-Saharan Africa scanned in routine care (Adewole et al., 2026). Single-institution releases that link MRI with molecular markers and expert segmentations support radiogenomic research and external testing (Reddy et al., 2026), and multi-cancer slide collections such as TCGA and CPTAC underpin most computational pathology work, including on glioblastoma (Saldanha et al., 2023). A recent benchmark deliberately includes cases in which pathology or reports are missing, so that methods can be tested under realistic incompleteness (Rivera et al., 2026). For human factors research, a large released dataset records radiologists reading with and without AI predictions and clinical history (Moehring et al., 2025).

Table 2. Public data resources referred to in this review.

Resource	Content	What it is used for	Main caveat
BraTS 2021 (Baid et al., 2021)	About 2,000 adult glioma cases, four MRI sequences, voxel-wise labels	Segmentation and radiogenomic benchmarking	Curated and pre-processed
BraTS-Africa (Adewole et al., 2026)	Routine-care glioma MRI from Sub-Saharan Africa	Testing transfer to low-resource imaging	Small, variable scanner quality
BraTS-PEDs (Kazerooni et al., 2024)	Multi-consortium paediatric brain tumour MRI	Paediatric segmentation	Rare tumour types, few cases per type
BraTS-MEN-RT (LaBella et al., 2026)	570 meningioma planning scans, 500 expert target volumes	Radiotherapy target delineation	Heterogeneous acquisition within and across sites
UTSW glioma dataset (Reddy et al., 2026)	Single-institution glioma MRI with molecular markers and segmentations	Radiogenomics and external testing	One institution

Resource	Content	What it is used for	Main caveat
TCGA and CPTAC slides (Saldanha et al., 2023)	Multi-cancer H&E whole-slide images with genetic profiling	Pathology pre-training, mutation prediction	Research cohorts, scanner and site effects
CoRe-BT (Rivera et al., 2026)	MRI for all cases, pathology and reports for some	Fusion under missing modalities	New; limited external use so far
Radiologist-AI collaboration dataset (Moehring et al., 2025)	Chest X-ray reads with and without AI and clinical history	Studying how readers use AI advice	Outside neuro-oncology

Two limitations of this landscape recur. First, benchmarks reward performance on their own test sets: a systematic review of 86 public neurosurgical datasets found that the baseline models released with them were often poorly validated externally and rarely reproducible (Chan et al., 2025). Second, the overlap metrics used to rank methods capture expert judgement only partly; in a study of how experts rate glioma segmentations, metric-rating correlations were moderate and weakest for the enhancing tumour, and network outputs were sometimes rated higher than the reference labels (Kofler et al., 2023). Challenge results also show that winning entries are usually ensembles of established architectures rather than new designs (Kazerooni et al., 2024; Ankomah et al., 2025). Federated benchmarking platforms, which send models to data held inside hospitals instead of the reverse, offer a way to test on representative clinical data without releasing it; one such platform evaluated 41 models across 32 sites (Karargyris et al., 2023).

4. Imaging: detecting and outlining brain tumours on MRI

4.1 Segmentation on curated data

Multi-parametric MRI (T1-weighted, contrast-enhanced T1, T2 and FLAIR) is the basis of pre-operative assessment, and voxel-wise segmentation of the enhancing tumour, tumour core and whole tumour is the most studied task. The BraTS 2021 benchmark provides about 2,000 annotated glioma cases with a standard preprocessing pipeline and evaluation metrics (Baid et al., 2021). On such data, the self-configuring nnU-Net adapts its preprocessing, architecture and training to each dataset and matched or exceeded specialised methods across many tasks (Isensee et al., 2021).

Transformer-based and hybrid designs followed. TransBTS inserted a transformer between a 3D convolutional encoder and decoder to model long-range context (Wang et al., 2021d); Swin UNETR reported five-fold mean Dice of 0.913 against 0.908 for nnU-Net on BraTS 2021 (Hatamizadeh et al., 2022); NestedFormer gave each sequence its own encoder and fused them with cross-modal attention, reaching an average Dice of 0.861 on BraTS 2020 with about 10

million parameters (Xing et al., 2022); and CKD-TransBTS added clinical knowledge about which sequence shows which sub-region (Lin et al., 2023). MedNeXt scaled convolutional networks with transformer-inspired blocks and outperformed seven baselines across four tasks (Roy et al., 2023).

The size of these gains should be read with caution. A re-evaluation by the nnU-Net authors, using identical preprocessing, matched compute budgets and five-fold cross-validation, found that many claimed improvements of transformer and state-space models over a well-configured U-Net did not hold, and that brain tumour segmentation benefited little from scaling (Isensee et al., 2024). A practical consequence is that a strong convolutional baseline should be part of every comparison.

4.2 Foundation and promptable models

General-purpose segmentation models have reached medical imaging. MedSAM adapted the Segment Anything model to medical images and outperformed the original model across tasks, often matching specialist models (Ma et al., 2024); MedSAM2 extended this to 3D volumes and video and reduced annotation effort in human-in-the-loop labelling (Ma et al., 2025); and adapter-based fine-tuning made such models usable with far fewer trainable parameters (Wu et al., 2025). BrainSegFounder, pre-trained on large collections of healthy-brain MRI, surpassed previous winning solutions on BraTS and stroke-lesion data, with performance increasing with model size and pre-training data (Cox et al., 2024). Parameter-efficient adaptation of large models to new structures with few labelled scans outperformed full fine-tuning at much lower compute (Silva-Rodríguez et al., 2023). These models are attractive where annotation capacity is limited, but they still require prompts or adaptation and their behaviour on small or ill-defined lesions needs local checking.

4.3 Incomplete protocols and real-world scans

Benchmark data contain all four sequences, but routine scans often do not. Several methods learn to segment when sequences are missing: mmFormer used modality-specific transformers and outperformed earlier approaches in all sequence combinations (Zhang et al., 2022a); M3AE used masked multimodal pre-training so that one model serves every combination (Liu et al., 2023); and adversarial co-training and latent-correlation methods transferred knowledge from complete to incomplete inputs (Wang et al., 2021b; Zhou et al., 2021). Aligning each sequence's features to a common reference distribution improved existing missing-modality methods by up to about 7.7 Dice points (Liu et al., 2024b).

Evidence from clinical data is more reassuring than these simulations suggest. On real-world glioma scans with missing sequences, nnU-Net achieved median Dice of 0.86 internally and 0.93 externally, within the range of human inter-rater agreement (Pemberton et al., 2023), and a multicentre metastasis study found that handling missing sequences preserved detection performance (Grovik et al., 2021). When models are moved between datasets, FLAIR transferred best, multi-sequence training helped further, and even a small amount of labelled target data gave rapid gains (Grande et al., 2026b). Differences between MRI sequences can cause larger performance drops than differences between hospitals (Zhang et al., 2025).

4.4 Metastases, paediatric tumours and meningiomas

Beyond adult glioma, deep learning detected brain metastases on post-contrast MRI with sensitivity above 93% in internal and external tests and raised reader sensitivity when used as an aid (Yin et al., 2022). In a randomised, multi-reader crossover study, AI assistance improved readers' metastasis segmentation (Dice 0.87 to 0.92) and saved 42% of contouring time (Luo et al., 2024). Multinational data showed that 2.5D and 3D models trade sensitivity against false positives differently across centres (Ottesen et al., 2022). For children, stepwise transfer learning from adult data reached expert-level segmentation, with AI contours judged acceptable more often than expert contours (Boyd et al., 2024); the BraTS paediatric challenge established a multi-consortium benchmark (Kazerooni et al., 2024); and a single-centre clinical model achieved Dice comparable to humans with up to 83% time reduction (Piffer et al., 2026). A dedicated meningioma radiotherapy dataset with 500 expert target volumes now supports planning research (LaBella et al., 2026).

4.5 Under-represented populations and low-resource imaging

Most public data come from well-resourced centres with modern scanners (Chan et al., 2025). The BraTS-Africa challenge curated routine-care glioma MRI from Sub-Saharan Africa, where scans are often acquired on older or lower-field systems and patients present later (Adewole et al., 2026). Pre-training on adult BraTS data and fine-tuning on the small African set reached an average Dice of 0.896 on unseen validation data (Hashmi et al., 2024), and the 2025 winning entry combined segmentation-aware augmentation with an ensemble of three architectures (Ankomah et al., 2025). These results show both that transfer works and that dedicated local data and adaptation are needed; South Asian populations remain largely absent from public brain tumour datasets (Chan et al., 2025).

4.6 Imaging biomarkers and molecular prediction

Because the current classification of central nervous system tumours integrates molecular markers, predicting them from MRI is an active area. A meta-analysis reported pooled sensitivity of 0.80 and specificity of 0.85 for IDH mutation and lower values for 1p/19q co-deletion, with significant heterogeneity (Farahani et al., 2026). Deep features outperformed handcrafted radiomics for IDH (AUC 0.915 against 0.830) (Zhang et al., 2023); perfusion imaging was the most informative sequence in glioblastoma (Pasquini et al., 2021); and foundation-based models generalised across external cohorts (Farahani et al., 2025a; Farahani et al., 2025b). Models trained on pooled public data and tested on local hospital patients predicted WHO 2021 molecular groups best when all sequences were used (Yang et al., 2025).

Two negative results deserve equal weight. For radiologically presumed lower-grade gliomas, deep learning was not clearly better than a simple clinical model in an international cohort (Gómez Vecchio et al., 2024), and an extensive evaluation of MGMT promoter methylation prediction from MRI found mean AUCs between about 0.55 and 0.63, close to chance (Saeed et al., 2023). Imaging-based molecular prediction should therefore be regarded as supportive evidence rather than a substitute for tissue testing.

A related caution applies to tumour classification on two-dimensional image collections assembled from public web sources. Studies on such data report accuracies of 90 to 99% (Alanazi et al., 2022; Karagoz et al., 2024), but the images are single, pre-selected slices rather than full clinical volumes, and results on very small sets are unstable: in one comparison a widely used network missed 11 of 31 tumours (Srinivas et al., 2022). These datasets are useful for teaching and quick baselines, not as evidence of clinical performance.

4.7 From research models to the reading room

Few studies describe deployment in routine practice. One neuroradiology group built a workflow that retrieves studies from the picture archiving system, runs segmentation and returns results to the reading environment within seconds; on its own patients the model reached a median Dice of 0.86, close to the 0.90 achieved on BraTS validation data (Aboian et al., 2022). Post-operative scans remain harder: a model for post-operative glioblastoma reached Dice of 0.81 for the whole tumour but only 0.64 to 0.71 for the resection cavity, necrosis and enhancing tumour, and adding the type of surgery and a cerebrospinal fluid segmentation reduced confusion between cavity, fluid and tumour (Holtzman Gazit et al., 2022). For meningioma radiotherapy, a probabilistic network slightly improved target delineation and its

uncertainty maps highlighted ambiguous boundaries for review (Wang et al., 2025). Imaging-based IDH prediction with an explicit confidence measure reached subject-wise AUC of 0.98 across several cohorts (Bangalore Yogananda et al., 2025). These studies suggest that integration into existing systems, and attention to the cases where the model is least reliable, are as important as headline accuracy.

5. Computational pathology

5.1 Learning from whole slides with weak labels

Digitised whole-slide images contain billions of pixels, while diagnoses are usually recorded only for the slide or patient. Attention-based multiple-instance learning (MIL) lets a slide-level label supervise attention over thousands of tiles (Ilse et al., 2018), and CLAM made this data-efficient and interpretable, with test AUCs of 0.95 to 0.99 on several cancer tasks (Lu et al., 2021). TransMIL modelled correlations between tiles (Shao et al., 2021), context-aware MIL down-weighted isolated noisy tiles (Fourkioti et al., 2023), and hierarchical self-supervised vision transformers scaled representation learning to gigapixel images (Chen et al., 2022b). A review concludes that MIL works because slide labels are routinely available, while interpretability, data efficiency and validation remain open (Gadermayr & Tschuchnig, 2022).

5.2 Foundation models and their limits

Self-supervised pre-training on large pathology collections produced general-purpose encoders. UNI reached balanced accuracies of 85.6% for IDH1 mutation and 88.3% for 12-class brain tumour subtyping (Chen et al., 2024c), and a visual-language model enabled zero-shot slide classification (Lu et al., 2024). Self-supervised encoders reduced labelling needs in several studies, matching supervised baselines with 5 to 40% of labels (Ye et al., 2024; Schirris et al., 2021), and DINO-trained transformers learned interpretable tissue concepts (Chen & Krishnan, 2022). A neuropathology-specific foundation model outperformed general models on brain tissue tasks (Verma et al., 2025).

Clinical benchmarks temper these gains. On real clinical slides, foundation models improved on ImageNet features but no single model was best on every task (Campanella et al., 2025); their embeddings retained strong hospital signatures that stain normalisation did not remove (Kömen et al., 2024); and the best adaptation strategy depended on the amount of local data (Lee et al., 2024b). Stain augmentation sampled from realistic colour distributions improved generalisation to other centres (Shen et al., 2022), and a lightweight self-supervised model was

more robust to scanner change than much larger foundation models, at a fraction of the compute (Shafique et al., 2025).

5.3 Glioma classification, grading and prognosis

For adult-type diffuse gliomas, a deep learning model trained on whole-slide images reached neuropathologist-level integrated classification, with AUCs of 0.90 to 0.99 across internal and two external test sets (Wang et al., 2023). Weakly supervised attention models distinguished glioma types and grades and inferred IDH status with AUC 0.95 in TCGA (Zuo et al., 2025); transfer learning on routine clinical-grade scans reached balanced accuracy of 0.87 for oligodendroglioma (Im et al., 2021); and a multicentre assessment showed accuracy falling from 0.79 internally to 0.73 across centres, a modest but real drop (Jin et al., 2023). For paediatric tumours, foundation features generalised better than ImageNet features but all extractors lost performance across sites (Tampu et al., 2026). Slide-based risk scores separated glioma patients into prognostic groups, alone or with IDH status and age (Chunduru et al., 2022; Luo et al., 2023), and self-supervised pipelines predicted many mutations across cancer types, including glioblastoma (Niehues et al., 2023; Saldanha et al., 2023).

Mutation prediction from slides is uneven. In a large multicentre comparison for colorectal cancer, the best pipeline predicted microsatellite instability with AUROC 0.94 and BRAF with 0.82, but KRAS, NRAS and PIK3CA only at 0.52 to 0.67, below clinical grade (Niehues et al., 2023). For low-grade glioma, combining multiple-instance learning with one-class classification improved prediction of several molecular markers over standard attention models (Fang et al., 2023). Integrating known molecular data with slide features improved glioma classification and grading (Pei et al., 2021). Small national cohorts can also be used effectively: with a neuropathologist-annotated WHO 2021 cohort, in-domain pre-training and retraining on model-labelled regions improved subtype classification over standard transfer learning (Despotovic et al., 2023). Frozen sections, which are the material available during surgery, differ from formalin-fixed slides; a contrastive approach that learned shared representations across both reached accuracy of 0.76 on fixed and 0.74 on frozen glioma slides, above single-domain training (Zhang et al., 2022b).

5.4 Fast tissue diagnosis during surgery

Several studies move diagnosis into the operating room. Stimulated Raman histology with deep learning predicted molecular groups of diffuse gliomas with mean accuracy of 93.3% within minutes (Hollon et al., 2023). Ultra-fast nanopore sequencing with a calibrated classifier

returned a methylation-based diagnosis within 40 minutes and abstained when uncertain (Vermeulen et al., 2023); a related approach classified 91 tumour classes with 94.5% accuracy from sparse methylation data (Brändl et al., 2025), and intra-operative methylation classification would have changed the surgical strategy in 12 of 20 cases (Djirackor et al., 2021). On frozen sections, an ensemble distinguished primary central nervous system lymphoma from glioma with external AUROC of 0.97, and combining AI with pathologists improved further (Zhang et al., 2024). An ensemble of pathology foundation models differentiated glioblastoma from its mimics with AUROC above 0.99 across centres (Zhao et al., 2025). A recent review summarises these advances in intra-operative AI (Goff et al., 2026).

6. Integrating imaging, tissue and molecular evidence

The integrated diagnosis of brain tumours combines several sources, and fusion methods aim to do the same. Early challenges on combined radiology and pathology classification showed that the best methods used both sources (Kurc et al., 2020), with weakly supervised fusion of slides and MRI performing competitively (Hsu et al., 2022) and a dual-network approach reaching balanced accuracy of 0.86 (Hamidinekoo et al., 2020). Pathomic Fusion combined histology and genomics and improved survival prediction for glioma over either source alone (Chen et al., 2022a). More recently, fusing pathology and radiology foundation models achieved AUC of 0.94 on a test set for WHO 2021 glioma subtyping, and remained effective even when trained on unpaired data (Saueressig et al., 2026). A new benchmark provides cases with MRI for all patients and pathology for some, so that fusion can be tested under realistic incompleteness (Rivera et al., 2026).

In that study the concordance index rose from 0.777 for a Cox model using grade and molecular subtype to 0.820 when histology and genomic features were fused (Chen et al., 2022a). A practical obstacle to fusion is that paired data are scarce: in the foundation-model study above, training on randomly paired radiology and pathology samples with the same label was as effective as patient-matched training and better than naive ensembling, which suggests that fusion can be learned from separately collected cohorts (Saueressig et al., 2026).

The consistent limitation is that most fusion studies assume every source is available and of good quality. In practice, histology lags imaging, molecular testing may be unaffordable, and slide quality varies. Robustness to missing or poor-quality sources is therefore as important as peak accuracy on complete cohorts.

7. Learning with limited labels and shifting data

Annotated local data are scarce in most hospitals, so label efficiency and robustness to distribution shift are central. Self-supervised contrastive pre-training of 3D encoders improved segmentation with few labels (Ali et al., 2021), and semi-supervised methods exploited unlabelled scans: uncertainty-guided consistency learning improved glioma segmentation with few labels (Zhang et al., 2021); a teacher-student curriculum raised Dice substantially as confident pseudo-labels were added (Liu et al., 2026); and self-training with tumour-aware deformations improved generalisation across tumour types (Grande et al., 2026a). Active learning combined with transfer reduced labelling by at least 40% for tumour classification (Hao et al., 2021), and contrastive anomaly detection identified tumours from only a few normal scans (Chin et al., 2024).

Few-shot segmentation that models only the target structure and treats everything else as anomalous can delineate new structures from a single labelled example (Hansen et al., 2022), and contrastive pre-training that also uses the few available pixel labels outperformed purely self-supervised and semi-supervised alternatives across label fractions (Hu et al., 2021a). Label efficiency and federation can also be combined: federated active learning selected the most informative cases for annotation across sites with domain shift (Chen et al., 2024b), and personalised prompts allowed federated training from weak labels such as scribbles and boxes to approach full supervision (Lin et al., 2024). For pathology, few-shot adaptation strategies were most useful for foundation models when local labels were very limited (Lee et al., 2024b).

For shift between scanners and sequences, unsupervised domain adaptation translated images between labelled and unlabelled sequences (He et al., 2021; d'Assier et al., 2022), causality-inspired augmentation improved single-source generalisation across centres and sequences (Ouyang et al., 2021), and privacy-preserving harmonisation separated anatomy from scanner style without sharing images (Galada et al., 2024). Synthetic tumours blended into public healthy brains allowed centres to share training data without sharing patients (Shi & Wang, 2025). Many studies in this area are preprints evaluated on public data; their behaviour on small, heterogeneous clinical cohorts is less certain.

8. Federated and privacy-preserving learning

Federated learning trains a shared model by exchanging parameters rather than data. Early brain tumour studies showed that federated training reached about 99% of the quality of a centrally trained model (Sheller et al., 2018; Sheller et al., 2020). The largest demonstration

trained a glioblastoma boundary model across 71 sites on six continents and improved the tumour core and whole-tumour delineation by 33% and 23% on out-of-sample data compared with a public initial model (Pati et al., 2022). The Federated Tumor Segmentation (FeTS) challenges benchmarked aggregation methods and exposed differences between institutions hidden by good average performance (Zenk et al., 2025; Linardos et al., 2025), and federated benchmarking platforms allow models to be evaluated on data that never leave the hospital (Karargyris et al., 2023). For paediatric brain tumours, an international federated platform achieved classification close to central training and improved segmentation at external sites (Lee et al., 2024a).

In that paediatric study the federated classifier reached an F1 of 0.856 against 0.877 for centralised training, a non-significant difference, whereas models trained at single sites performed poorly (Lee et al., 2024a). The FeTS challenges add two lessons: selective client sampling and adaptive aggregation improved both performance and training efficiency, and median Dice above 0.9 at most institutions coexisted with much weaker results at a few (Zenk et al., 2025); the 2024 winner used a controller-based aggregation that also converged fastest (Linardos et al., 2025). Large pre-trained segmentation models can themselves be trained federatedly, with adapter-only training reducing communication substantially (Liu et al., 2024a), and federation can work even when sites hold different imaging modalities (Parekh et al., 2021). Not every comparison favours federation, however: in one MRI classification study an ensemble reached 96.7% accuracy when trained centrally but 91.1% when federated (Islam et al., 2022).

Heterogeneity between sites is the main technical difficulty. FedProx and FedBN address differences in clients and in feature distributions (Li et al., 2020; Li et al., 2021); harmonisation, domain generalisation and personalisation methods reduce drift between heterogeneous medical sites (Jiang et al., 2022; Liu et al., 2021; Jiang et al., 2023; Wang et al., 2022); and modality-specific encoders handle sites with different MRI sequences (Dai et al., 2024). Yet an extensive benchmark found that plain federated averaging already performs close to central training for brain tumour segmentation, with other methods giving small gains (Manthe et al., 2024), and a general benchmark found no algorithm best on every medical dataset (Zhou et al., 2025). Real deployments add practical barriers: a nationwide pathology federation in Germany had to operate behind hospital firewalls (Babendererde et al., 2026), and a three-hospital study found performance varied by site and most improvement occurred in the first 10 to 15 rounds (Hindawi et al., 2025). Federated pathology on gigapixel slides is feasible with differential

privacy (Lu et al., 2020; Adnan et al., 2022), and swarm learning removes the central server altogether (Saldanha et al., 2022).

Federation is not automatically private. Shared gradients can be inverted to reconstruct medical images under default settings (Das et al., 2024), differential privacy costs accuracy as the privacy budget tightens (Singh et al., 2020; Nampalle et al., 2023), and robust aggregation is needed when a site sends faulty or malicious updates (Karami et al., 2026). Reviews highlight that heterogeneity also complicates the estimation of model reliability in federated settings (Koutsoubis et al., 2024; Guan et al., 2023).

9. Reliability of model outputs

Deep networks can be confidently wrong, especially on scans that differ from their training data, so estimating the reliability of each output has become part of model development. The QU-BraTS challenge evaluated voxel-wise uncertainty for tumour segmentation and showed that good uncertainty estimates allow uncertain voxels to be filtered with a gain in Dice, while poor estimates are penalised (Mehta et al., 2022). Conformal prediction has been applied to glioma segmentation, with uncertain cases having markedly lower Dice than certain ones (0.71 against 0.90) (Elyassirad et al., 2025). In radiogenomics, predictions with the lowest uncertainty were more accurate than the rest (83% against 48% for EGFR amplification) (Hu et al., 2021b), and automatic quality control that flags failed segmentations raised the average quality of the accepted cases (Williams et al., 2021). Evaluations of uncertainty in multi-task glioma models found that uncertainty behaved differently for each task (Rojas et al., 2026). These results suggest that reliability estimates are useful for quality control, but their value depends on how they are validated and presented to users, which is discussed next.

10. Clinical translation

10.1 Surgery, post-operative assessment and planning

AI is reaching the surgical stages of care. Automated segmentation on early post-operative MRI classified the extent of glioblastoma resection comparably to experts (Luque et al., 2024), and a validated post-operative model reached Dice of 0.81 for residual enhancing tumour (Cepeda et al., 2024). A perioperative tool, however, determined the correct extent of resection in only 44% of cases, showing that post-operative assessment remains difficult (Zanier et al., 2023), and cleaner, consistent training labels significantly improved tumour volume segmentation for radiotherapy (Hochreuter et al., 2024). Deep learning also segmented gliomas

on intra-operative ultrasound with Dice of 0.90 on hold-out data but lower on other cohorts (Cepeda et al., 2025).

Pre-operative planning relies on tractography and functional mapping. Advanced tractography reconstructed the corticospinal tract more completely than standard tensor methods (Sheng et al., 2021; Zhylka et al., 2021); tractography corrected for brain shift was highly accurate for subcortical resections (Ille et al., 2021); and the distance between the resection and the motor tract was associated with preserved motor function (Eibl et al., 2024). Machine learning has generated safe surgical trajectories consistent with expert choices in a proof of concept (Dundar et al., 2022), networks can estimate MRI-ultrasound registration error during surgery (Salari et al., 2023), and virtual and augmented reality changed how surgeons identified structures and chose their approach (Zawy Alsofy et al., 2021; Mofatteh et al., 2022; Gómez Amarillo et al., 2023). These studies mostly report geometric accuracy; evidence on surgeon acceptability and on effect on outcomes is still limited.

Language and visual pathways add further complexity. Navigated tractography of language pathways supported maximal safe resection near eloquent cortex and identified zones where damage tends to be permanent (Salvati et al., 2021), and a review of connectome-guided surgery describes a shift from removing the lesion to protecting brain networks, while noting tractography errors, cost and lack of standardisation (Patel et al., 2026). Intra-operative imaging, fluorescence and optical techniques each improve precision but add interpretation burden, and reviews expect AI to help integrate them (Bin-Alamer et al., 2023; Goff et al., 2026). Augmented and virtual reality have been used safely during awake craniotomy to support mapping (Mofatteh et al., 2022), and low-cost browser- and smartphone-based tools are being developed for settings without navigation systems (Ho et al., 2024; Asadi et al., 2024).

10.2 Clinicians working with AI

Reader studies show that AI assistance can help. Assistance for brain metastasis segmentation improved accuracy and saved time, with the largest benefit for residents (Luo et al., 2024); AI for glioma grading raised neuroradiologists' accuracy from 82.5% to about 88% (Jin et al., 2024); and a meta-analysis across imaging found that AI reduced reading time by 27% and reading volume by up to 62%, with sensitivity maintained or improved (Chen et al., 2024a). In pathology, AI assistance improved Gleason grading beyond both pathologists alone and the AI

alone (Bulten et al., 2021), increased agreement in PD-L1 scoring across institutions (Wang et al., 2021c), and reduced the use of extra stains and second opinions (Eloy et al., 2023).

The benefits are uneven. The effect of AI assistance varied widely between radiologists and could not be predicted from experience (Yu et al., 2024). In liver cancer classification, assistance improved pathologists' accuracy when the model was right and reduced it significantly when it was wrong, with no overall gain (Kiani et al., 2020). False-positive AI marks raised suspicion and changed decisions in cerebral aneurysm reading, especially for less experienced readers (Kim et al., 2025a); automation bias occurred in 7% of pathology decisions, and time pressure increased its severity (Rosbach et al., 2024; Rosbach et al., 2026). Collaboration works best when clinicians weigh the AI's reliability case by case: endoscopists followed AI more when it was correct than when it was wrong, and the team outperformed both (Reverberi et al., 2022). A survey of European radiologists found that most users considered AI results reliable but few observed a reduction in workload (European Society of Radiology (ESR), 2022).

Deployment studies explain part of this gap. In rural primary-care clinics, clinicians often ignored or closed an AI decision-support system because of time pressure, the effort of entering data and suggestions they did not find convincing (Wang et al., 2021a). Interviews about time-constrained decisions highlight the need to align AI output with guidelines, explain disagreements and fit the consultation (Jacobs et al., 2021), and tools that let clinicians refine and question an imperfect algorithm increased both its usefulness and their trust without loss of accuracy (Cai et al., 2019). Large language models show similar patterns: a model optimised for differential diagnosis produced a correct diagnosis in its top ten more often than unassisted clinicians (59.1% against 33.6%), and clinicians assisted by it did better than those using search engines alone (51.7% against 44.4%) (McDuff et al., 2025).

Explanations are not a simple remedy. Sixteen popular heatmap methods failed clinical requirements on multi-sequence brain MRI (Jin et al., 2022; Jin et al., 2021); explanations increased trust in only about half of clinical studies and could raise acceptance of wrong advice (Rosenbacke et al., 2024); and showing high confidence scores increased trust while reducing accuracy (Rezaeian et al., 2025). Cognitive forcing designs reduced over-reliance at a cost in user satisfaction (Bucinca et al., 2021), and presenting uncertainty as frequencies helped users adjust their reliance (Cao et al., 2024). A recent scoping review concluded that team benefits

depend on task fit, workflow integration, training and calibrated trust, and that patient outcomes are rarely measured (Strong et al., 2026).

Acceptance is a precondition for any of this. Neurosurgeons worldwide reported strong interest in AI but limited knowledge of it (Boaro et al., 2025); surgical teams were most supportive of real-time hazard alerts and operative planning (Layard Horsfall et al., 2021); and trust, transparency, liability and workflow fit shape acceptance beyond classic technology-acceptance factors (Scipion et al., 2025; Asan et al., 2020). Reporting guidelines for early clinical evaluation now require these human factors to be measured (Vasey et al., 2022).

10.3 Language models in neuro-oncology

Large language models (LLMs) are being tested as decision aids. Assisted by an LLM, readers' accuracy for brain MRI differential diagnosis rose from 46.5% to 61.4% (Kim et al., 2025b); knowledge-enhanced LLMs classified gliomas from radiologist-scored features about as well as experienced radiologists and raised junior readers' AUC from 0.77 to 0.83 (Li et al., 2025); and LLM help mainly benefited trainees reading brain tumour cases (Song et al., 2026). In neuropathology, zero-shot models often failed, while grounding them in the current WHO classification chapters raised correct subtyping to 90% (Hewitt et al., 2024). For tumour boards, an LLM agreed with neuro-oncology radiotherapy decisions in 76% of cases but sometimes omitted indicated treatment (Tini et al., 2025), and LLM-written board summaries were rated as plausible as human ones (Goldberg et al., 2025). Reviews stress that such tools must support rather than replace clinicians (Lawson McLean et al., 2024). These studies are small and mostly retrospective, but they show that grounding in current guidelines is essential.

On structured clinical reasoning tasks the picture is similar. On 305 neurology board questions, a reasoning model answered 90.9% correctly on its own and higher when given retrieval of reference material and access to tools (Sorka et al., 2025). In a study of brain MRI differential diagnosis, assistance from a language model improved readers' accuracy but required careful checking of its suggestions (Kim et al., 2025b), and effects were largest for trainees (Song et al., 2026). Across these studies, grounding the model in current guidelines and classification systems, and keeping a clinician responsible for the final decision, were consistently emphasised.

11. Evaluating AI in neuro-oncology

11.1 Validation and comparison

Several studies in this review show how easily technical claims are overstated. Under identical preprocessing and compute budgets, many reported architecture gains disappeared (Isensee et al., 2024); plain federated averaging matched more complex aggregation for brain tumour segmentation (Manthe et al., 2024); and conventional clinical models matched deep learning for IDH prediction in a realistic cohort (Gómez Vecchio et al., 2024). Good practice that follows from this evidence includes external test sets from different institutions, comparison with strong simple baselines, confidence intervals for every metric, and reporting of failure cases rather than averages alone. Because overlap metrics only partly reflect expert judgement (Kofler et al., 2023), segmentation studies benefit from combining overlap and boundary metrics with expert rating of a sample of cases.

11.2 Reader studies

The reader studies cited above use a small number of designs that can be reused. Multi-reader, multi-case studies in which the same readers interpret the same cases with and without AI, separated by a washout period, isolate the effect of assistance (Kim et al., 2025a; Kiani et al., 2020); randomised crossover designs control for learning and order effects (Luo et al., 2024); and stratifying readers by experience shows who benefits (Luo et al., 2024; Song et al., 2026). Deliberately including cases in which the AI is wrong is essential, because average benefit can hide harm on those cases (Kiani et al., 2020; Kim et al., 2025a). Simulated studies before deployment allow interfaces to be refined without risk to patients (Kunz et al., 2025; Määttä et al., 2023).

11.3 Workload, usability and trust

Efficiency is measured by reading time and reading volume (Chen et al., 2024a), and perceived workload by the NASA Task Load Index, for which free online tools exist (Pandian & Suleri, 2020); in a randomised crossover study, a decision-support interface nearly halved paramedics' reported mental demand (Lu et al., 2026). Usability is commonly measured with the System Usability Scale, on which a well-designed intensive-care prototype scored 87.5 (Kunz et al., 2025), and expert heuristic evaluation can find problems with user control and error prevention before clinicians test a tool (Cho et al., 2022). Expert inspection and end-user testing find different problems, so combining them gives a fuller picture (Rizvi et al., 2025). Whether a

tool should interrupt the user or wait to be consulted depends on the user's role and on the cost of missed advice (Blecker et al., 2019).

Trust needs careful handling. Trust questionnaires do not capture whether people rely on AI appropriately, that is, follow it when it is right and not when it is wrong, and the constructs used to measure reliance are still fragmented (Raees & Papangelis, 2026). A decision-theoretic framework separates losses caused by following the AI at the wrong times from losses caused by being unable to tell good from bad advice (Guo et al., 2024). People who overestimate their own competence under-use good advice (He et al., 2023), while learning from feedback and examples improves appropriate reliance over time (Schemmer et al., 2023). Adapting explanations to the user's current level of trust reduced inappropriate reliance by up to 38% in one study (Srinivasan & Thomason, 2025).

11.4 Reporting standards

Reporting guidelines now cover early clinical evaluation of decision-support systems, with explicit items on human factors, the interaction between clinician and system, and safety in real use (Vasey et al., 2022). Studies of AI in neuro-oncology that follow such guidance, report how the system was presented to clinicians and analyse the cases where it failed will be more useful to the field than further retrospective accuracy studies.

12. Discussion

12.1 What is established

Four conclusions are well supported. Segmentation of brain tumours on curated multi-sequence MRI is technically mature, and simple, well-configured baselines remain competitive (Isensee et al., 2021; Isensee et al., 2024). Slide-level computational pathology, now supported by foundation encoders, reaches specialist-level accuracy for major glioma types (Wang et al., 2023; Chen et al., 2024c). Federated learning is a proven route to multi-institutional training without moving patient data (Pati et al., 2022; Manthe et al., 2024). And AI assistance improves average efficiency and often accuracy in reader studies (Chen et al., 2024a; Luo et al., 2024).

12.2 What remains open

Representativeness. Public datasets leave many populations unrepresented, and baseline models released with them are often poorly validated externally (Chan et al., 2025). Dedicated benchmarks for under-represented populations show that transfer works but is not sufficient (Adewole et al., 2026; Hashmi et al., 2024).

Robustness to real practice. Performance falls across centres in pathology (Jin et al., 2023; Tampu et al., 2026), foundation models carry hospital signatures (Kömen et al., 2024), sequence differences cause larger drops than site differences (Zhang et al., 2025), and fusion methods usually assume complete inputs (Saueressig et al., 2026). Post-operative and intra-operative tasks remain harder than pre-operative segmentation (Zanier et al., 2023; Cepeda et al., 2025).

Honest evaluation. Several claimed advances disappear under rigorous comparison (Isensee et al., 2024; Manthe et al., 2024), imaging-based molecular prediction can fail to beat simple clinical models (Gómez Vecchio et al., 2024; Saeed et al., 2023), and common overlap metrics only partly reflect expert judgement of segmentation quality (Kofler et al., 2023). External validation, appropriate baselines and confidence intervals should be standard.

Human factors. Average benefits hide large individual differences and harm from incorrect AI output (Yu et al., 2024; Kiani et al., 2020). Explanations and confidence displays can increase reliance without improving accuracy (Rosenbacke et al., 2024; Rezaeian et al., 2025). Evaluations rarely follow clinicians beyond short reading sessions or measure patient outcomes (Strong et al., 2026).

Equity and accountability. Models trained on unrepresentative data may perform worse for the populations least represented in them, and reviews of AI in neurosurgical oncology identify bias, generalisability and data security as the main concerns (Baker et al., 2024; Chan et al., 2025). Clinicians' trust depends not only on accuracy but on transparency, perceived risk and who carries legal responsibility when the system errs (Asan et al., 2020; Jones et al., 2023; Schulz et al., 2025), and generative tools raise a further risk that clinicians' own diagnostic skills erode over time (Al-Anezi, 2026). Federated learning reduces data movement but does not by itself guarantee privacy (Das et al., 2024).

12.3 Directions for research

The evidence points to several directions. First, local and regional datasets from under-represented populations, with careful annotation protocols, are needed for both development and validation (Hochreuter et al., 2024; Adewole et al., 2026). Second, methods should be tested on the incompleteness of routine care, including missing sequences, delayed pathology and varied scanners, rather than on simulated gaps (Pemberton et al., 2023; Rivera et al., 2026). Third, label-efficient and federated learning should be studied together in the realistic regime of a few small, heterogeneous hospitals, with attention to privacy risks and site-level

performance (Hindawi et al., 2025; Das et al., 2024). Fourth, evaluations should follow reporting guidelines for early clinical studies, measure workload, usability and trust alongside accuracy, and examine how clinicians respond when the AI is wrong (Vasey et al., 2022; Kim et al., 2025a). Finally, surgical planning tools should be evaluated for their effect on surgeons' decisions and on patient outcomes, not only for geometric accuracy (Zawy Alsofy et al., 2021; Eibl et al., 2024).

12.4 Implications for low-resource settings

Several findings are directly useful where data, specialists and computing are limited. Strong, openly available baselines such as nnU-Net perform well on real clinical data, including scans with missing sequences (Pemberton et al., 2023), and are a sensible starting point before more complex models. Transfer from large public datasets followed by fine-tuning on small local cohorts works for under-represented populations and for children (Hashmi et al., 2024; Boyd et al., 2024), while a small amount of locally labelled data gives rapid gains (Grande et al., 2026b). Lightweight models can be more robust to scanner change than large foundation models at a fraction of the compute (Shafique et al., 2025), and adapters make large models usable without full retraining (Silva-Rodríguez et al., 2023; Wu et al., 2025). Federated learning has been deployed behind hospital firewalls (Babendererde et al., 2026), and synthetic data sharing and privacy-preserving harmonisation offer further options when sites cannot join live training (Shi & Wang, 2025; Galada et al., 2024). In every case, local validation on the population to be served remains necessary, because public benchmark performance does not guarantee performance elsewhere (Chan et al., 2025; Adewole et al., 2026).

12.5 Limitations of this review

This review is narrative and does not include a formal quality assessment or meta-analysis. It was restricted to open-access full texts, which may under-represent some journals, and many recent studies are preprints that have not yet been peer reviewed. Quantitative results come from different datasets and protocols and should not be compared directly.

13. Conclusion

AI for brain tumours has matured from isolated models into a broad set of tools spanning imaging, pathology and surgery. The main obstacles are no longer architectural. They are the representativeness of data, robustness to the incompleteness of routine care, honest external validation, and understanding of how clinicians and AI perform together. Addressing these will

matter most in settings where specialists and data are scarce, which is where AI could add the most value.

Declarations

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